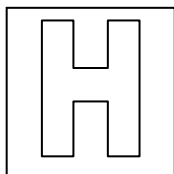


Name: Answers	Index No.:	CT Group: 10
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PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CHEMISTRY

9647/03

Paper 3 Free Response

12 September 2011

2 hours

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** out of the five questions.

A Data Booklet is provided.

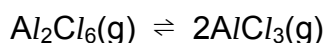
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

Answer any **four** questions

- 1 (a) Solid aluminium chloride sublimes at 180°C. In the vapour phase, an equilibrium is established between aluminium chloride and its dimer.



At 400°C and 1 atm, 0.84 g of the equilibrium gaseous mixture occupies 268 cm³.

- (i) Calculate the average relative molecular mass of the gaseous mixture, giving your answer to 1 decimal place.

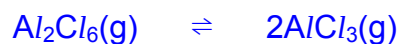
$$M_r \text{ of gaseous mixture} = \frac{0.84 \times 8.31 \times (400 + 273)}{101000 \times 268 \times 10^{-6}} = 173.6$$

- (ii) Determine the degree of dissociation of Al_2Cl_6 , α , using the formula:

$$\alpha = \left(\frac{M_r(\text{Al}_2\text{Cl}_6) - \text{average } M_r}{\text{average } M_r} \right)$$

$$\alpha = \left(\frac{267 - 173.6}{173.6} \right) = 0.538$$

- (iii) Hence, calculate the value of the equilibrium constant, K_p , for the dissociation of Al_2Cl_6 at 400°C.



Initial amount/ mol	1	0
Δ in amount / mol	- 0.538	+1.076
Eqm amount / mol	0.462	1.076

Total amount = 1.538 mol

$$\text{Partial pressure of } \text{Al}_2\text{Cl}_6 = \frac{0.462}{1.538} \times 1 = 0.300 \text{ atm}$$

$$\text{Partial pressure of } \text{AlCl}_3 = 1 - 0.300 = 0.700 \text{ atm}$$

$$K_p = \frac{0.700^2}{0.300} = 1.63 \text{ atm}$$

- (iv) State and explain the effect of increasing the temperature on the average M_r of the equilibrium mixture.

The dissociation of Al_2Cl_6 is an endothermic process as it involves the breaking of two covalent / dative bonds. According to Le Chatelier's principle, increasing the temperature favours an endothermic reaction. Thus, position of equilibrium shifts to the right. There will be a greater proportion of $AlCl_3$ in the reaction mixture, causing the average M_r of the reaction mixture to decrease.

[7]

- (b) (i) Explain what is meant by the term *bond energy*.

Bond energy is the energy absorbed to break one mole of covalent bonds between two atoms in the gaseous state.

Or

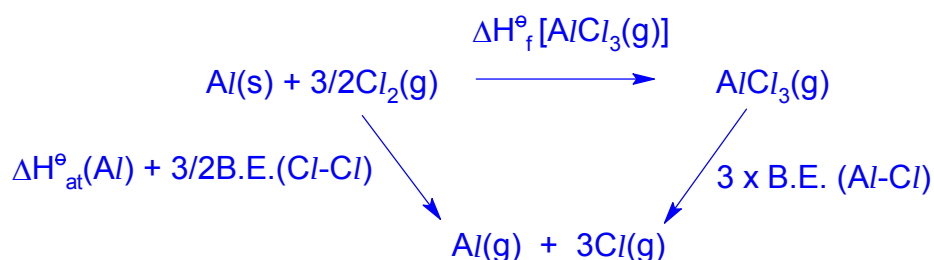
Bond energy is the energy absorbed to break one mole of covalent bonds in a gaseous molecule.

- (ii) By means of an energy cycle, calculate the average bond energy of the $Al-Cl$ bond using the following data and information from the *Data Booklet*.

Standard enthalpy change of formation of $AlCl_3(g) = -585 \text{ kJ mol}^{-1}$

Standard enthalpy change of atomisation of $Al = +314 \text{ kJ mol}^{-1}$

[4]

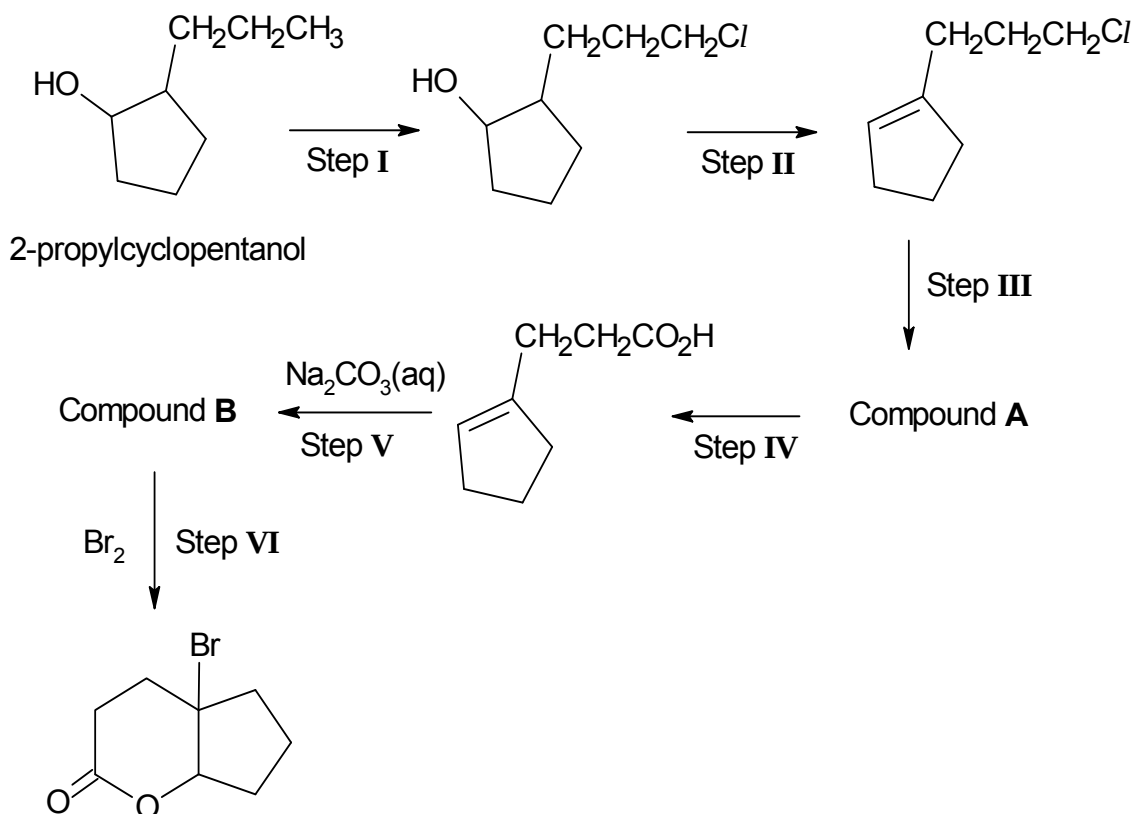


Applying Hess' Law,

$$3 \times B.E.(Al-Cl) = -(-585) + (314) + 3/2(244)$$

$$B.E.(Al-Cl) = +422 \text{ kJ mol}^{-1}$$

- (c) One of the important uses of halogens is in the preparation of halogen-containing organic compound. One particular synthesis is given below.



- (i) In step I, a sample of 2-propylcyclopentanol and limited amount of chlorine are mixed together and irradiated with ultraviolet light.

The mechanism involves two propagation steps.

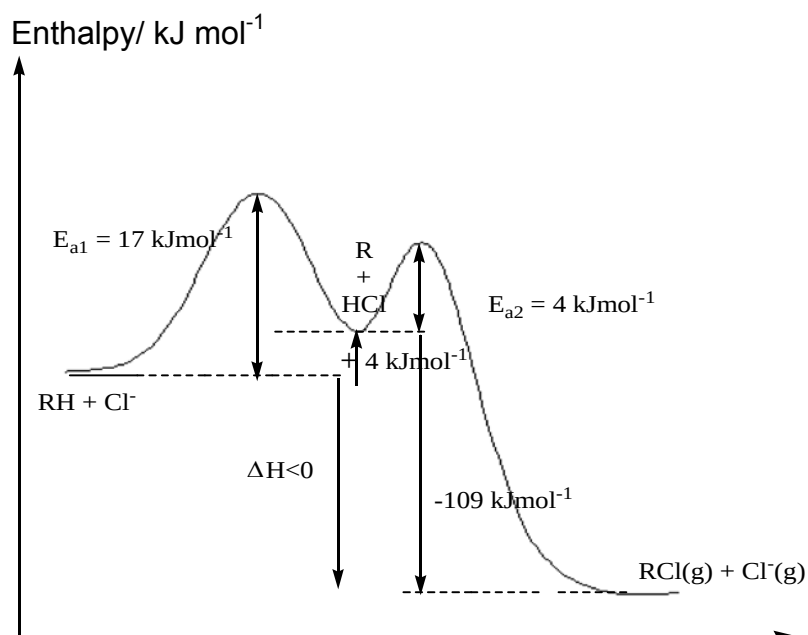
(RH represents 2-propylcyclopentanol)



The respective enthalpy changes of reaction and activation energies for the two propagation steps are given below:

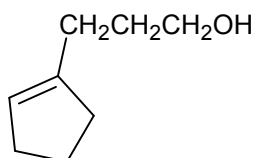
Step	$\Delta H / \text{kJ mol}^{-1}$	$E_a / \text{kJ mol}^{-1}$
Propagation step 1	+4	17
Propagation step 2	-109	4

Using the information provided, sketch the combined energy profile diagram for the two propagation steps of the chlorination of 2-propylcyclopentanol, showing all relevant energy changes and the reaction intermediates.

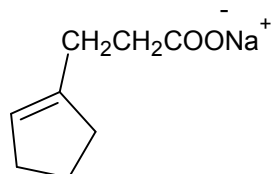


- (ii) Give the structural formulae of compounds **A** and **B**.

Compound A:



Compound B:



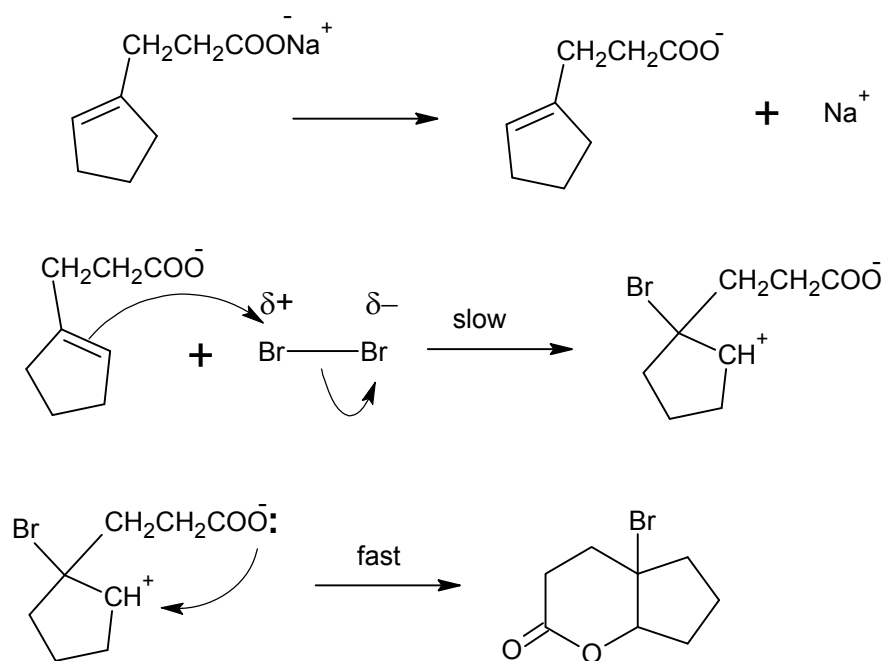
- (iii) State the reagents and conditions that are required to carry out the reactions in steps **II**, **III** and **IV**.

Step **II**: excess concentrated H_2SO_4 , 170°C

Step **III**: NaOH(aq) , heat

Step **IV**: $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4(\text{aq})$, heat

- (iv) Given that step VI involves electrophilic addition, propose a suitable mechanism for the reaction.



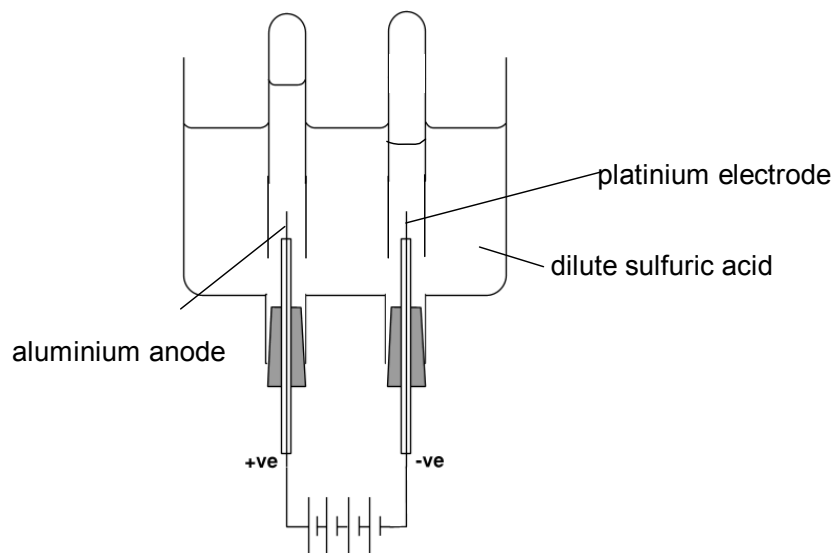
[9]

[Total: 20]

- 2 (a) The anodised aluminium layer is grown by passing a direct current through dilute sulfuric acid, with the aluminium object serving as the anode.

When current is passed through the circuit, the oxygen liberated at the anode will react with aluminium to create a build-up of aluminium oxide.

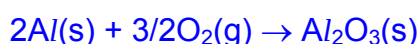
3.0 A of current is passed through the circuit and the volume of the oxygen collected at the anode after 10 minutes is 40 cm³.



- (i) Write ion-electron equations to account for the liberation of the two gases at each electrode.



- (ii) Write an equation to show the formation of aluminium oxide.



- (iii) Calculate the volume of oxygen you would expect to be produced at the anode using inert electrodes at room temperature and pressure.

$Q = 3.0 \times 600 = 1800 \text{ C}$

No. of moles of electrons = $1800/96500 = 0.01865 \text{ moles}$

No. of moles of O₂ produced = $0.01865/4 = 4.663 \times 10^{-3} \text{ moles}$

Volume of O₂ produced = $4.663 \times 10^{-3} \times 24 = 0.112 \text{ dm}^3$

- (iv) Using your answer in (a)(iii), calculate the mass of aluminium oxide produced at the anode.

Volume of O₂ reacted with aluminium = $111.9 - 40 = 71.9 \text{ cm}^3$

No. of moles of O₂ reacted = $0.0719/24 = 3.329 \times 10^{-3} \text{ moles}$

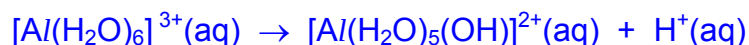
Mass of Al₂O₃ produced = $3.329 \times 10^{-3} \times 2/3 \times 102 = 0.226 \text{ g}$

- (b) (i) Aluminum chloride is very soluble in water and its solution has a pH of approximately 3. With aid of equations, account for the observations.

$AlCl_3$ first dissolves to form $[Al(H_2O)_6]^{3+}$ ion.



Al^{3+} has high charge density / polarising power (or Al^{3+} polarises surrounding coordinated H_2O molecules and weakens the O-H bonds, causing them to release H^+ ions).



- (ii) What will be observed when aqueous sodium hydroxide is added gradually to aqueous aluminum chloride until it is present in excess? Give the formula of the aluminium-containing species formed in this reaction.

White precipitate of $Al(OH)_3$ will be observed. White precipitate dissolves in excess NaOH giving colourless solution of $[Al(OH)_4]^- Na^+$ or $[Al(OH)_4]^-$.

[4]

- (c) A recently discovered organic compound **L** possesses a pleasant floral scent and is therefore being used as a component of many essential oils and perfumes. A molecule of **L** has the molecular formula, $C_{10}H_{18}O$.

A sample of **L** has the ability to rotate plane-polarised light.

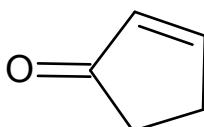
No reaction occurs between **L** and 2,4-dinitrophenylhydrazine. However, **L** produces **M**, $C_{10}H_{22}O$, with hydrogen with nickel catalyst.

No change is observed when **L** is warmed with acidified $K_2Cr_2O_7$.

Heating **L** with Al_2O_3 , produces a mixture of two products. **N**, with molecular formula $C_{10}H_{16}$, is present as the major product.

N reacts with hot concentrated acidified $KMnO_4$ to produce propanone and **P**, $C_5H_6O_5$. A colourless gas which forms a white precipitate with limewater is liberated.

P can also be produced by the reaction of the compound below with hot concentrated acidified $KMnO_4$.

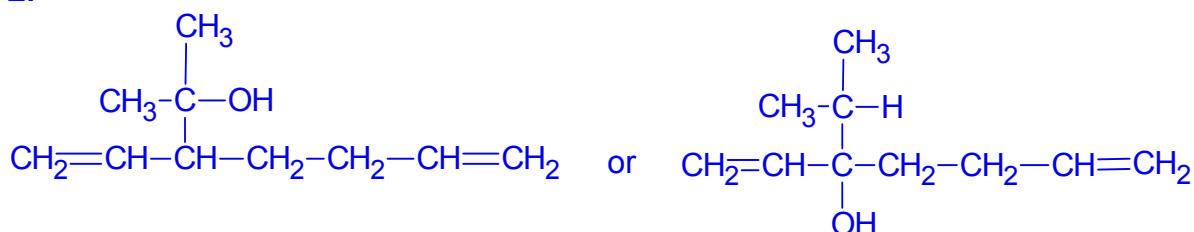


Deduce the structures of compounds **L**, **M**, **N**, and **P**. Explain the chemistry of the reactions described.

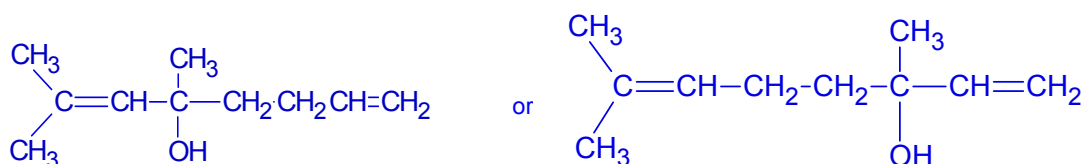
[9]

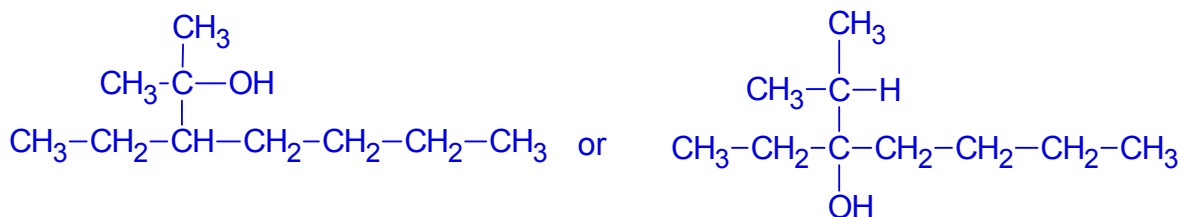
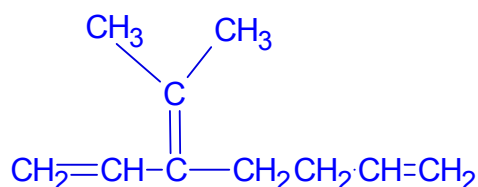
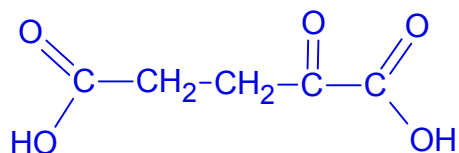
Information/ Type of reaction	Deductions: functional group
L rotates plane-polarised light.	L exhibits optical isomerism and contains at least 1 <u>chiral carbon</u>
No reaction occurs between L and 2,4-dinitrophenylhydrazine.	L has <u>no aldehyde or ketone functional group</u> .
L produces M, C ₁₀ H ₂₂ O, when it undergoes <u>reduction</u> [$\frac{1}{2}$] with hydrogen with nickel catalyst	L is unsaturated. Additional of 4 H atoms indicate presence of <u>2 C=C bonds</u> .
L <u>does not undergo oxidation</u> with acidified potassium dichromate(VI).	L contains a <u>3° alcohol group</u> .
L undergoes <u>elimination</u> when heated with aluminium oxide to form N as major product.	L contains <u>alcohol</u> that undergoes elimination. N contains an additional <u>C=C bond</u> upon elimination.
N undergoes <u>oxidation</u> with hot concentrated acidified KMnO ₄ to produce propanone and P, C ₅ H ₆ O ₅ . CO ₂ gas which gives a white precipitate with limewater is liberated.	Presence of <u>2 terminal C=C bond</u> due to 2 moles of CO ₂ produced during oxidative cleavage. (loss of 2 C from N to propanone & P).
P can also be produced by <u>oxidation</u> of the compound shown above with concentrated acidified KMnO ₄ and heating.	-

L:



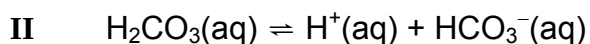
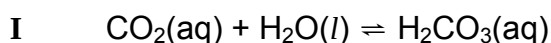
(Note: The 2 structures below are not accepted as they will lead to formation of a minor product following Zaitsev's rule)



M:**N:****P:**

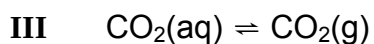
- 3 (a) Human blood is slightly basic, with a normal pH of approximately 7.4. Its pH must be maintained for the effective transport of oxygen.

When CO_2 is dissolved in the blood plasma, a series of equilibria is established.



Equilibrium **II** has a K_a value of $7.90 \times 10^{-7} \text{ mol dm}^{-3}$ and is responsible for buffering the pH of the blood plasma at 7.4.

In the lungs, excess dissolved $\text{CO}_2(\text{aq})$ is exhaled as $\text{CO}_2(\text{g})$ according to the following equilibrium.



During heavy exercise, lactic acid is released into the blood and is buffered by the blood plasma. The buffering action of the blood plasma will eventually lead to an increased breathing rate.

- (i) State what is meant by a *buffer solution*.

A buffer solution is one that resists changes in pH when a small amount of acid or alkali is added, or upon dilution.

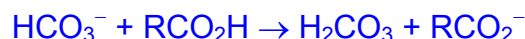
- (ii) Use the data above to determine the $[\text{HCO}_3^-(\text{aq})]/[\text{H}_2\text{CO}_3(\text{aq})]$ ratio in the blood plasma.

$$7.90 \times 10^{-7} = [\text{H}^+][\text{HCO}_3^-]/[\text{H}_2\text{CO}_3]$$

$$7.90 \times 10^{-7} = 10^{-7.4} [\text{HCO}_3^-]/[\text{H}_2\text{CO}_3]$$

$$[\text{HCO}_3^-]/[\text{H}_2\text{CO}_3] = \underline{19.8}$$

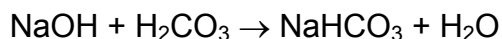
- (iii) Write an equation to show how lactic acid is removed from the body. Use RCO_2H to represent lactic acid.



- (iv) Hence, by considering the effect on equilibria I and III, explain why the process of removing lactic acid causes the breathing rate to increase.

As more H_2CO_3 is formed, position of equilibrium I shifts left. This increases $[\text{CO}_2(\text{aq})]$, and in turn causes the position of equilibrium III to shift right, i.e. breathing rate increases.

- (v) A student wishes to simulate the blood plasma buffer by mixing 25.0 cm^3 of $0.20 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$ solution with $0.30 \text{ mol dm}^{-3} \text{ NaOH}$ solution. The neutralisation that takes place is represented by



Using your answer in (a)(ii), calculate the volume of NaOH that should be added.

[8]

$$\begin{aligned} \text{Let the volume of NaOH added be } V \text{ cm}^3. \\ \text{Initial amount of H}_2\text{CO}_3 &= 0.20 \times 25.0 \times 10^{-3} \\ &= 5.0 \times 10^{-3} \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Amt of H}_2\text{CO}_3 \text{ left after neutralisation} &= 5.0 \times 10^{-3} - 0.30 V \times 10^{-3} \\ &= 0.005 - 3 \times 10^{-4} V \text{ mol} \end{aligned}$$

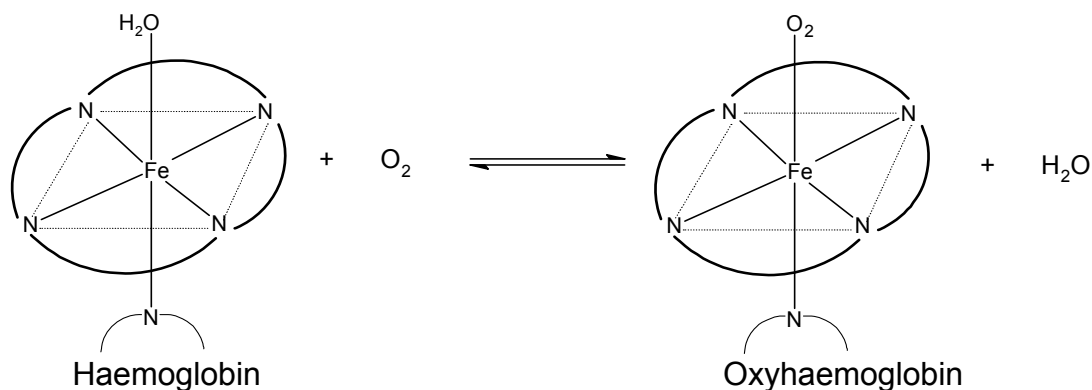
$$\begin{aligned} \text{Amt of HCO}_3^- \text{ formed} &= \text{Amt of NaOH used} = 3 \times 10^{-4} V \text{ mol} \\ 3 \times 10^{-4} V / [0.005 - 3 \times 10^{-4} V] &= 19.8 \\ V &= \underline{15.9 \text{ cm}^3} \end{aligned}$$

[8]

- (b) Haemoglobin (Hb) is an oxygen carrier protein that circulates around the body within red blood cells.

Each Hb molecule contains four subunits ($\alpha_2\beta_2$), which each contains a haem group responsible for binding oxygen.

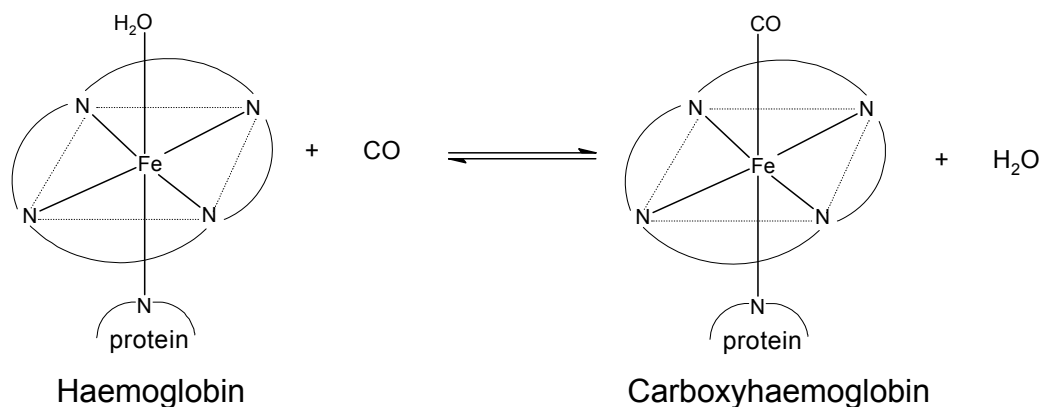
A haem is an iron chelate of a tetrapyrrole macrocycle. The iron ion is situated in the centre of the planar macrocycle, and is the site of oxygen binding. Haem can be represented schematically as shown.



- (i) Describe the bonding around the iron ion.

Dative covalent bonds/ coordinate bonds [1] are formed between the N centres of the tetrapyrrole/ O₂ and the iron ion, where N/ O donates a lone pair into vacant 3d orbitals of Fe.

- (ii) In the blood of smokers, another form of Hb, carboxyhaemoglobin, is also common.



CO is found to have 245 times more affinity for haemoglobin than O₂. Explain how CO gives rise to toxicity/ poisoning in the blood of smokers.

CO is a stronger ligand than H₂O (or O₂),

CO binds strongly and irreversibly to Fe, unlike H₂O or O₂.

If compared with H₂O:

CO undergoes ligand exchange, displacing H₂O from haem to form

carboxyhaemoglobin.

If compare with O_2 :

CO prevents O_2 from binding to Fe OR:

O_2 is unable to displace CO OR:

CO displaces H_2O from Fe more readily than O_2 is able to do so.

Carboxyhaemoglobin is a more stable complex than oxyhaemoglobin.

- (iii) Explain why CO is unable to displace the tetrapyrrole macrocycle from the iron centre.

[4]

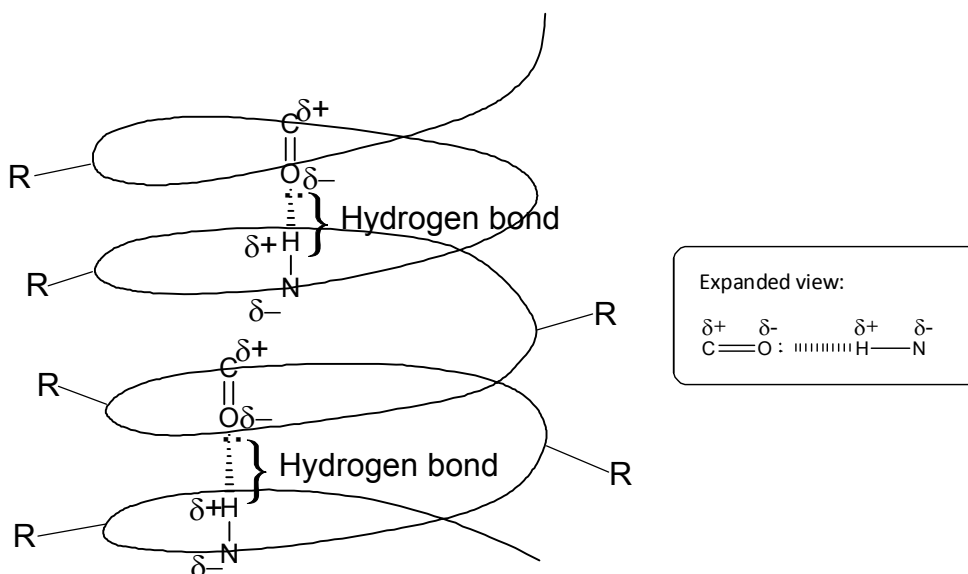
The tetrapyrrole macrocycle is a tetradentate ligand, which can form multiple dative bonds to the Fe centre.

Or

The tetrapyrrole macrocycle exhibits the chelate effect when binding to Fe centre, where replacement of the tetrapyrrole macrocycle with 4 CO ligands will lead to a negative entropy change ($\Delta S < 0$), and is hence energetically feasible.

- (c) The secondary structure of haemoglobin consists of segments of α -helices. Repeated folding of the secondary structure gives rise to a highly globular protein subunit. Four globular protein subunits ($\alpha_2\beta_2$) assemble to form haemoglobin.

- (i) Describe the α -helix structure with the aid of a suitable diagram.



Marking points:

- α -helix
- at least 1 hydrogen bond shown with:
 - (1) labeled dipole moments on N and H
 - (2) labeled hydrogen bond
 - (3) lone pair on O shown

Hydrogen bond formed between --NH of one peptide linkage, and --CO of another peptide linkage on the polypeptide chain.

- (ii) The amino acids which are on the exterior of haemoglobin are likely to interact strongly with water molecules.

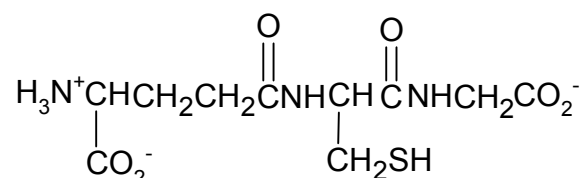
amino acid	formula of side chain (R in $\text{RCH}(\text{NH}_2)\text{CO}_2\text{H}$)
serine	$\text{--CH}_2\text{OH}$
aspartic acid	$\text{--CH}_2\text{CO}_2\text{H}$
valine	$\text{--CH}(\text{CH}_3)_2$
lysine	$\text{--CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$
phenylalanine	$\text{--CH}_2\text{C}_6\text{H}_5$

Identify two amino acids from the above table that are likely to have this property. Explain your choice.

serine, lysine, aspartic acid.

The --OH and --NH_2 groups present in the polar R groups of Ser/ Asp and Lys respectively are able to form hydrogen bonds with surrounding H_2O molecules, accounting for its hydrophilicity.

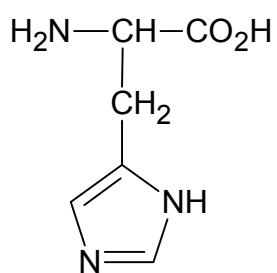
- (iii) A mutated tripeptide segment of an α -polypeptide chain of haemoglobin was obtained via partial hydrolysis.



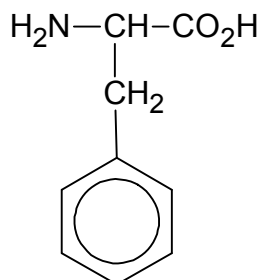
What is unusual about the peptide bond formed by the N-terminal amino acid in this tripeptide?

The peptide bond is formed using the --COOH of the R group of the N-terminal amino acid, instead of the $\alpha\text{--COOH}$ group.

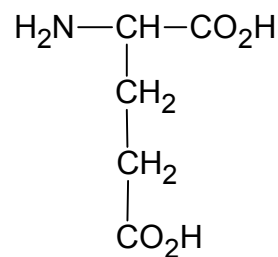
- (iv) Another tripeptide, his-phe-glu, from a β -polypeptide chain was hydrolysed. The resulting solution was added to an excess of a buffer solution of **pH 5.5** and placed at the centre of the plate. A potential difference was then applied across the plate.



his
pI 7.6
pK_a (2° amine): 4.0;
pK_a (N of C=N): 6.0

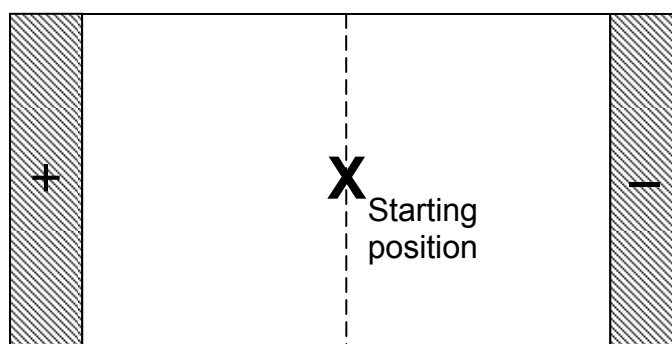


phe
5.5

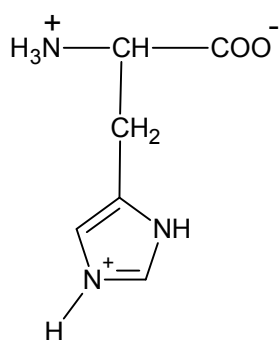


glu
3.1

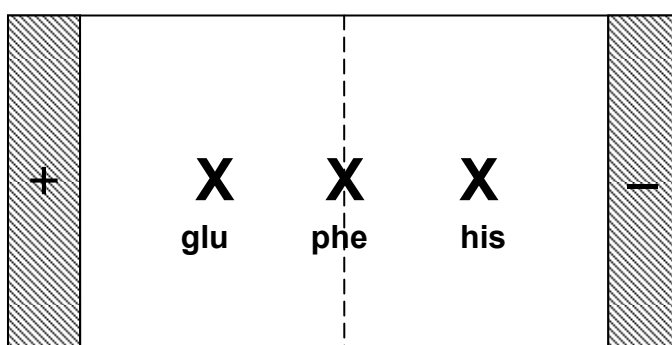
- (I) Draw the structural formula of the resulting histidine at this pH.
(II) Copy the diagram shown below onto your writing paper and indicate the relative positions of the three amino acids on the diagram.



[8]



his



- 4 Silver and lead form a series of halides of general formula AgX and PbX_2 . The halides of these two cations are sparingly soluble in water at 25°C .

Salt	Solubility product / K_{sp}
AgCl	$2.00 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$
PbCl_2	$1.70 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$
PbBr_2	$6.60 \times 10^{-6} \text{ mol}^3 \text{ dm}^{-9}$

- (a) A solution of 100 cm^3 of $0.120 \text{ mol dm}^{-3} \text{ AgNO}_3(\text{aq})$ was accidentally mixed with 50 cm^3 of $0.120 \text{ mol dm}^{-3} \text{ Pb}(\text{NO}_3)_2(\text{aq})$. Two methods were suggested to separate the cations.

- (i) In the first method, 0.180 g of solid NaCl was added to the mixture. Show, by means of calculation, how the two cations can be separated by selective precipitation.

$$\begin{aligned} \text{No. of moles of NaCl added} &= 0.180 / (23.0 + 35.5) \\ &= 3.077 \times 10^{-3} \text{ mol} \end{aligned}$$

$$\begin{aligned} [\text{NaCl}]_{\text{after dissolving}} &= 3.077 \times 10^{-3} / (150/1000) \\ &= 0.0205 \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} [\text{Ag}^+(\text{aq})]_{\text{after mixing}} &= \text{no. of moles of Ag}^+(\text{aq}) \text{ before mixing} \div \text{total volume} \\ &= (100/1000 \times 0.120) \div (150/1000) \\ &= 0.0800 \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} [\text{Pb}^{2+}(\text{aq})]_{\text{after mixing}} &= \text{no. of moles of Pb}^{2+}(\text{aq}) \text{ before mixing} \div \text{total volume} \\ &= (50/1000 \times 0.120) \div (150/1000) \\ &= 0.0400 \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} \text{Ionic product of AgCl} &= [\text{Ag}^+(\text{aq})]_{\text{after mixing}} [\text{Cl}^-(\text{aq})]_{\text{after dissolving}} \\ &= 0.0800 \times 0.0205 \\ &= 1.64 \times 10^{-3} \text{ mol}^2 \text{ dm}^{-6} > K_{\text{sp}}(\text{AgCl}) \end{aligned}$$

Hence, $\text{AgCl}(\text{s})$ will form

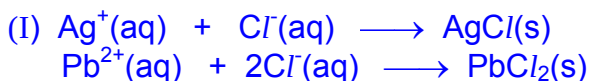
$$\begin{aligned} \text{Ionic product of PbCl}_2 &= [\text{Pb}^{2+}(\text{aq})]_{\text{after mixing}} [\text{Cl}^-(\text{aq})]_{\text{after dissolving}}^2 \\ &= 0.0400 \times (0.0205)^2 \\ &= 1.68 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9} < K_{\text{sp}}(\text{PbCl}_2) \end{aligned}$$

Hence, no $\text{PbCl}_2(\text{s})$ will form.

- (ii) The alternative method of separating the two metal ions is to add **excess** aqueous NaCl to the mixture in (a) followed by excess $\text{NH}_3(\text{aq})$ and then filtration.

Construct ionic equations for

- (I) the reactions between the mixture in (a) and excess NaCl(aq).
 (II) the reaction that takes place when excess NH₃(aq) is added.



[7]

- (b) When a precipitate is formed, $\Delta G_{\text{ppt}}^\theta$, in J mol⁻¹, is given by the following expression.

$$\Delta G_{\text{ppt}}^\theta = 2.303 RT \log K_{\text{sp}}$$

- (i) Use the data above to calculate $\Delta G_{\text{ppt}}^\theta$ for lead(II) bromide.

$$\begin{aligned} \Delta G_{\text{ppt}}^\theta(\text{PbBr}_2) &= 2.303 RT \log K_{\text{sp}}(\text{PbBr}_2) \\ &= 2.303 \times 8.31 \times (273+25) \log (6.60 \times 10^{-6}) \\ &= -29544 \\ &= -29500 \text{ J mol}^{-1} \end{aligned}$$

- (ii) The standard enthalpy change of precipitation of lead(II) bromide is -279 kJ mol⁻¹. Hence, calculate $\Delta S_{\text{ppt}}^\theta$, in J mol⁻¹ K⁻¹, for lead(II) bromide at 298K.

$$\begin{aligned} \Delta G_{\text{ppt}}^\theta &= \Delta H_{\text{ppt}}^\theta - T\Delta S_{\text{ppt}}^\theta \\ -29544 &= (-279 \times 10^3) - (298) \Delta S_{\text{ppt}}^\theta \\ \Delta S_{\text{ppt}}^\theta &= -837 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$$

- (iii) Explain the significance of the sign of your answer in (ii).

Since $\Delta S_{\text{ppt}}^\theta < 0$, the formation of PbBr₂(s) results a decrease in entropy of the system, i.e. from a more disordered aqueous solution to a less disordered solid formed.

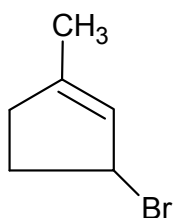
- (iv) Assuming ΔS_{ppt} does not change with temperature, state the effect of increasing temperature on ΔG_{ppt} and hence solubility of lead(II) bromide.

Increasing temperature causes magnitude of $T\Delta S$ to rise above magnitude of ΔH . This causes ΔG to become positive. Precipitation is non-spontaneous, indicating that solubility of PbBr₂ increases with increasing temperature.

[4]

- (c) Student A from Pioneer Junior College found a bottle of a liquid substance that is labeled, C₆H₉Br. He suggested that the unknown has the following structure.

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- (i) Plan **two** chemical reactions that can positively confirm the suggested structure. Include your observations and deductions for each test.

Test 1:

Add NaOH(aq), heat. Cool the mixture. Acidify with HNO₃(aq). Add AgNO₃(aq)

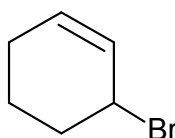
Cream precipitate of AgBr formed. It contains bromoalkane.

Test 2:

Add Br₂(aq).

Decolourisation of reddish brown solution. It contains C=C double bond.

- (ii) Student **B** from the same college thinks otherwise and proposed that the unknown liquid has the structure below.



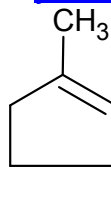
Propose a chemical test which would enable you to distinguish between them. Include your observations.

Test:

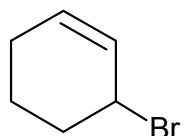
Add KMnO₄, NaOH(aq), I₂(aq), warm

Observation:

If yellow precipitate of CHI₃ & brown ppt of MnO₂ observed confirms

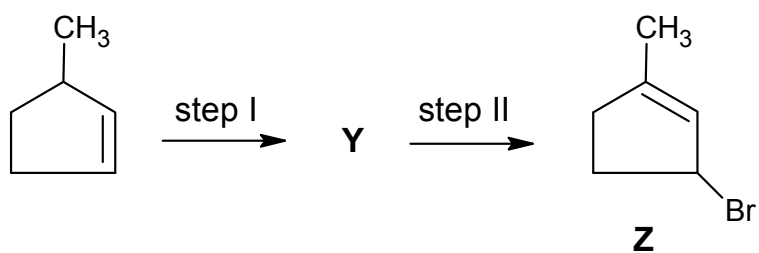


present as it contains CH₃CO group.

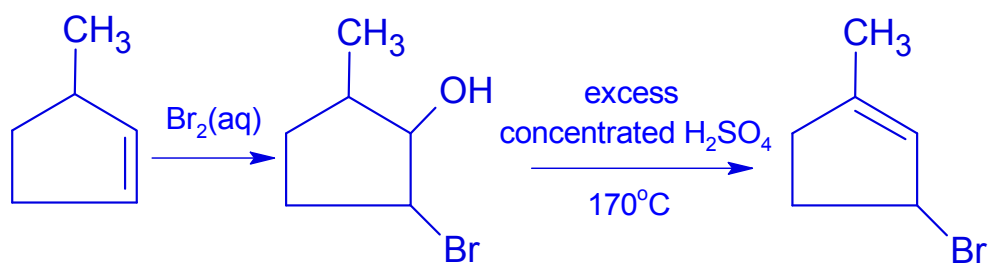


No yellow precipitate observed in

(iii) Compound **Z** can be formed by the reaction sequence below.



Suggest the structural formula of compound **Y**. State the reagents and conditions in each of the steps.



[9]

[Total: 20]

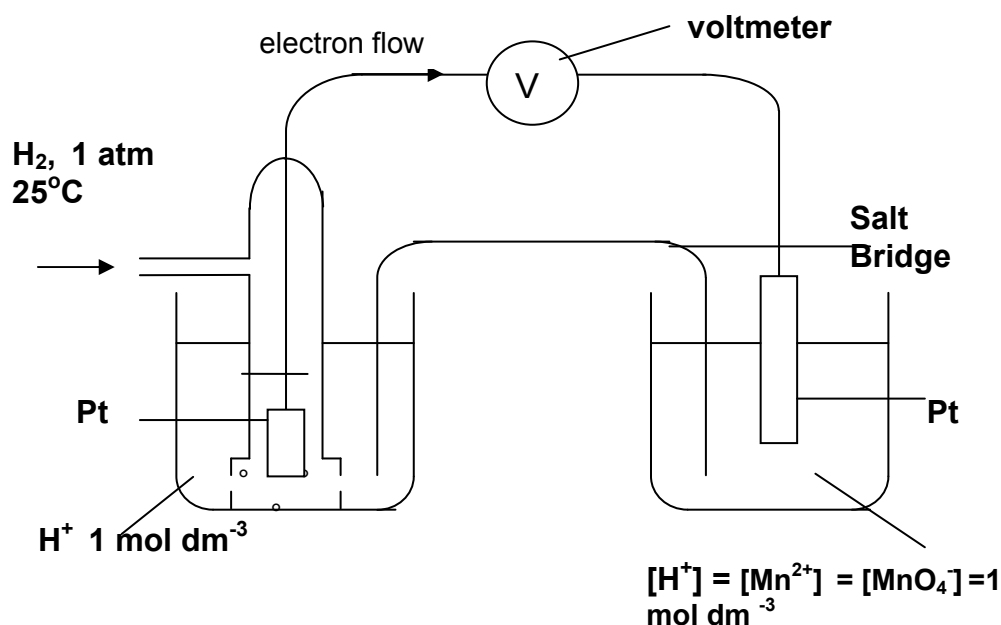
- 5 (a) The standard redox potential of manganate(VII) ions is given below:



- (i) What do you understand by the term *standard electrode potential*?

The potential difference between a standard hydrogen electrode and a half cell under standard conditions of 25°C, 1 atm and concentration of any ion is 1 mol dm⁻³.

- (ii) Draw a fully labeled sketch of how you would determine the standard electrode potential of the $\text{MnO}_4^-(\text{aq}) \mid \text{Mn}^{2+}(\text{aq})$ electrode system.

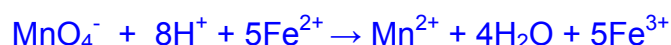


[3]

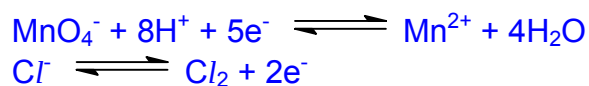
- (b) Potassium manganate(VII) is also frequently used in titrimetric analysis due to its strong oxidising property.

A student was tasked to determine the amount of iron(II) salts present in a nutritional supplement pill. He decided to carry out a titration with a standard solution of acidified KMnO_4 that he prepared.

- (i) Write a balanced equation for the reaction of MnO_4^- and Fe^{2+} in acid solution.



- (ii) The acid that was used to prepare the standard solution was sulfuric acid. Using appropriate data from the *Data Booklet*, explain why hydrochloric acid *could not* be used.



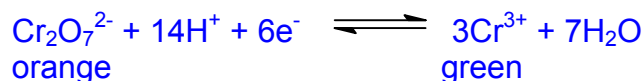
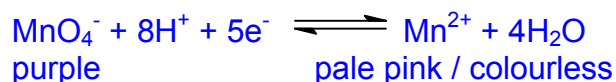
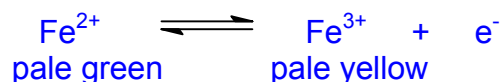
$$E^\ominus_{\text{red}} = +1.52 \text{ V}$$

$$E^\ominus_{\text{ox}} = -1.36 \text{ V}$$

$$E^\ominus_{\text{cell}} = +0.16 \text{ V}$$

Cl^- ions present would be oxidised to Cl_2 by MnO_4^- . This would cause the titre value to be higher than expected.

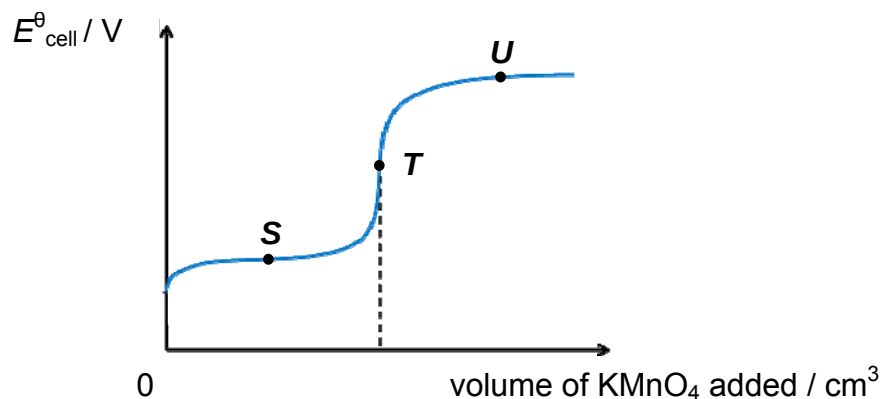
- (iii) Suggest why potassium manganate(VII) rather than potassium dichromate(VI) was used in the titration.



The end point colour change will be more distinct / sharp colour change and hence can be detected more easily when $\text{KMnO}_4(\text{aq})$ is used.

[4]

One such nutritional supplement pill containing Fe^{2+} was dissolved in a volume of dilute sulfuric acid and titrated against $0.0200 \text{ mol dm}^{-3} \text{ KMnO}_4$. The $E^\ominus_{\text{cell}}$ was measured against a standard hydrogen electrode and the following graph was obtained.



- (iv) Using suitable data listed in the *Data Booklet*, suggest approximate $E^\ominus$ values, in volts, for the points **S** and **U** on the graph.

At point **S**, the mixture contains Fe^{2+} , Fe^{3+} and Mn^{2+} .
Hence, equilibrium exists between Fe^{2+} and Fe^{3+} ions.

$$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+} \quad E^\ominus = +0.77 \text{ V}$$

At point **U**, the mixture contains Fe^{3+} , Mn^{2+} and MnO_4^- .
Hence, equilibrium exists between Mn^{2+} and MnO_4^- ions.



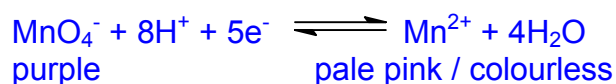
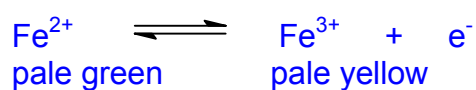
- (v) State an important assumption that was made to estimate the answers in part (b)(iv).

1) At point **S**, $[\text{Fe}^{3+}] = [\text{Fe}^{2+}]$ and at point **U**, $[\text{MnO}_4^-] = [\text{Mn}^{2+}]$.

or

2) The temperature of the mixture remains constant throughout the entire titration process.

- (vi) State the colour of the solution at point **T**.

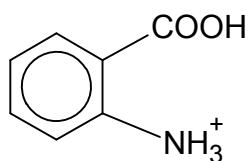


Colour of solution at point **T** is orange.

[4]

- (c) Compound **H** has the molecular formula, $\text{C}_9\text{H}_9\text{NO}$.

- **H** is insoluble in aqueous sodium hydroxide and hydrochloric acid at room temperature.
- When **H** is heated with acidified potassium manganate(VII), a compound with the following structure is obtained.



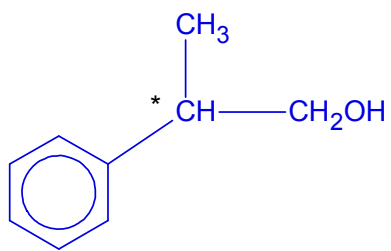
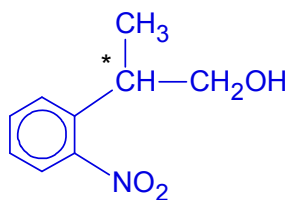
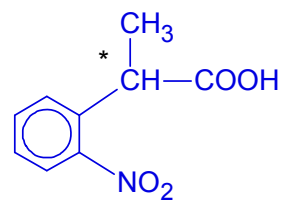
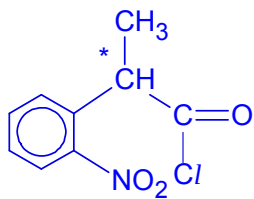
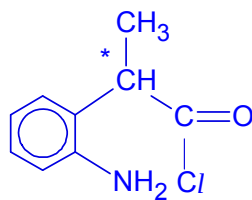
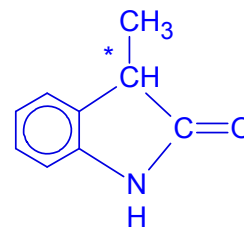
Compound **H** can be prepared by a series of reaction starting from optically active compound **J**, $\text{C}_9\text{H}_{12}\text{O}$.

- Compound **J**, $\text{C}_9\text{H}_{12}\text{O}$, reacts with concentrated nitric acid and sulfuric acid to give **K**, $\text{C}_9\text{H}_{11}\text{NO}_3$.
- Treating compound **K** with acidified potassium dichromate(VI) gives compound **L**, $\text{C}_9\text{H}_9\text{NO}_4$.
- Compound **L** is then reacted with phosphorus pentachloride to give **M**, $\text{C}_9\text{H}_8\text{NO}_3\text{Cl}$.

- Reduction of compound **M** under suitable conditions yields **N**, $C_9H_{10}NOCl$, which undergoes immediate reaction by eliminating a molecule of hydrogen chloride to give **H**.

Deduce the structures of **H**, **J**, **K**, **L**, **M** and **N**. explain the chemistry of the reactions described.

H is insoluble in aqueous sodium hydroxide and hydrochloric acid at room temperature.	H neutral; H is a amide
When H is heated with acidified potassium manganate(VII) under suitable condition, a compound with the given structure is obtained.	H undergo acidic hydrolysis and oxidation H has is 1,2-position substituted
Compound H can be prepared by reacting optically active compound J , $C_9H_{12}O$, with concentrated nitric acid and sulfuric acid to give K , $C_9H_{11}NO_3$.	J has a chiral centre J is not phenol, J is aldehyde/alcohol J undergo electrophilic substitution; $-NO_2$ group substituted at relative postion 2
Compound K reacts with acidified potassium dichromate(VI) to give compound L , $C_9H_9NO_4$.	K undergoes oxidation to give carboxylic acid K is a 1° alcohol L is carboxylic acid
Compound L is then reacted with phosphorus pentachloride to give M , $C_9H_8NO_3Cl$.	L undergoes nucleophilic substitution to give M M is acyl chloride.
Under suitable conditions, compound M is reduced to give N , $C_9H_{10}NOCl$, which undergoes immediate reaction by eliminating a molecule of hydrogen chloride to give H .	$-NO_2$ reduced to $-NH_2$ - Acyl chloride and $-NH_2$ react to eliminate a molecule of HCl, giving cyclic compound H amide present in H.

**J**, $\text{C}_9\text{H}_{12}\text{O}$ **K**, $\text{C}_9\text{H}_{11}\text{NO}_3$ **L**, $\text{C}_9\text{H}_9\text{NO}_4$ **M**, $\text{C}_9\text{H}_8\text{NO}_3\text{Cl}$ **N**, $\text{C}_9\text{H}_{10}\text{NOCl}$ **H**, $\text{C}_9\text{H}_9\text{NO}$

[9]
[20]

End of Paper